1. TITLE OF THE IDEA

Apex PdM: Township-Scale Asset Health & Air-Conditioning Predictive Maintenance Platform

A low-cost sensor pod that feels, hears and measures machines, so township teams fix small problems before they become big breakdowns

LIVE WORKING RIG: https://washer-pdm.pages.dev **Stage:** Working prototype (TRL 4–5) with live data online

10	4	20×+	~RM 170
REAL WASH SESSIONS ARCHIVED	SENSORS WORKING IN SYNC	VIBRATION JUMP, IDLE → SPIN	PILOT KIT COST, OFF THE SHELF

2. PROBLEM STATEMENT

What is the issue? Air-conditioners, chillers, water pumps and big appliances are usually fixed only **after they break**. Before that, a worn bearing, damper or seal slowly makes the machine shake more, sound different and **work harder, using more electricity**. Nobody notices until the machine stops. By then, a **small part worth tens of ringgit** has turned into a **new motor or compressor worth thousands**, plus days without cooling.

Who is affected?

- **Township Communal & Commercial Spaces:** Community **mosques, suraus, kuils (temples), clubhouses, and retail shopping malls** operating many commercial AC compressors daily. Sudden breakdown leaves congregations and shoppers in unbearable tropical heat.
- **Township Facility Management (FM) Teams:** Technicians check machines one by one, by hand, and still cannot see wear happening inside until something fails.
- **Township Residents:** Households facing surprise repair bills, noisy machines and higher electricity bills.

Why is it important to solve this problem now? Electricity keeps getting more expensive, and townships are aiming for lower emissions and less waste. A machine that is wearing out uses more power and gets thrown away sooner. Catching the problem early means **a cheap repair instead of a full replacement**, lower bills and less scrap going to landfill.

3. PROPOSED SOLUTION

How does your solution address the problem? Apex PdM is a small sensor pod that sits on the outside of a machine. It records **vibration, sound and power together**, learns what *normal* looks like for that machine, and gives an **anomaly score and a simple verdict for every run** (normal / check it). A full health score and remaining-life estimate are our next step, once we have recorded real faults.

1. Vibration (MPU-6050 @ 200 Hz): 3-axis sensor that feels shaking, unbalance and loose or worn parts.	**2. Sound (INMP441 @ 16 kHz):** Digital microphone that hears grinding, squealing and rattling.
3. Power (Tapo P110 @ 1 Hz): Smart plug that measures how hard the motor is working, in watts.	**4. Machine State:** Context from SmartThings (wash, rinse, spin); comparing vibration and sound within the same phase is planned.

What technology / product is involved?

- **Dual-Core ESP32:** One core reads the sensors in real time, the other handles Wi-Fi and cloud upload, so neither slows the other down. Every 5 seconds the pod turns raw signals into **32 signal features**.
- **Keeps Logging Through Wi-Fi Drops:** The pod stores about 4 minutes of data and sends it once the connection is back.
- **Live Web Dashboard:** Anyone can watch the rig live today. A township map and phone alerts are the next build.

4. TARGET MARKET / BENEFICIARIES

Specific Industry or Sector: Commercial Facility Management (FM), Communal Places of Worship, Retail Real Estate, and Smart Sustainable Townships (e.g., Sime Darby Property's City of Elmina, Bandar Bukit Raja).

- **Communal Places of Worship:** Mosques, Suraus, and Kuils (temples) relying on central air-conditioning split units.
- **Commercial Centers & Clubhouses:** Community gyms, function halls, and retail malls (e.g., Elmina Lakeside Mall).
- **Local Maintenance Contractors (MSMEs):** Technicians who get told which machine needs a look, instead of rushing to emergency calls.

Urban/Rural Focus: High-density master-planned urban townships and commercial property clusters.

5. UNIQUENESS AND INNOVATION

- **Low Cost:** Industrial vibration sensors cost thousands of ringgit each, so only big factories use them. We spent **about RM 131 in parts for the sensor node** (about RM 170 for a full pilot kit with smart plug) — cheap enough for every AC unit in a mosque or mall.
- **Fewer False Alarms by Design:** One sensor alone gets fooled — a normal fast spin looks like a fault. We check vibration, sound and power **together**, and plan to check against what the machine is doing, before raising an alert.
- **Sees Energy Waste Too:** Because we measure power directly, the system can compare each run's power use against that machine's own normal and show when it starts working harder.
- **Easy to Fit:** Mounts on the outside and plugs in — no rewiring and nothing opened up.

6. PROTOTYPE FEASIBILITY, EMPIRICAL EVIDENCE & THE EXPANSION PLAN

Why we tested on a washing machine: A front-load washer is one of the toughest machines to monitor. It shakes hard, the load moves around, water sloshes, and the speed keeps changing from slow tumbling to a fast spin. If our pod gets clean, clear data here, a steady-running AC compressor or pump is an easier job. **What we measured** in our latest 66-minute cycle: vibration jumped **over 20×** from idle to spin; the main vibration frequency climbed smoothly to **19.9 Hz (about 1,200 rpm)** as the drum sped up; and the washer's sound in the higher bands was **36–90× louder** than the room's background noise, even with a compressor and water pump running nearby.

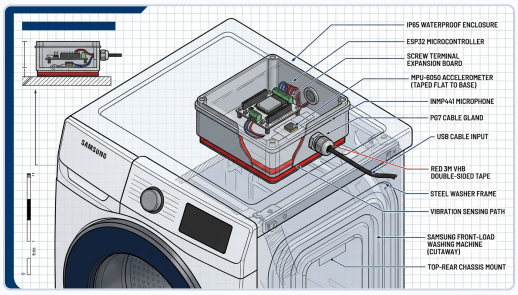


Figure 1: Sensor pod layout
- **Hardware:** ESP32 dual-core + MPU-6050 (200 Hz) + INMP441 mic (16 kHz) + Tapo P110 smart plug (1 Hz).
- **Data in Hand: 10 real wash sessions archived** and **7,414 rows** of training-ready data, 32 signal features per 5-second window (firmware-07 window record).
- **Model Validation:** Leave-one-cycle-out test on 8 healthy cycles: 0 falsely flagged (95% upper bound 31%); a simulated doubling of sound was caught in 7 of 8.
- **Live Global Dashboard:** Operational worldwide at **https://washer-pdm.pages.dev**.

From washer to township: Fans, pumps and compressors wear out the same way — worn parts cause more shaking, then more noise, then more power use, then a breakdown. The same pod and software can move onto **AC units and pumps in township mosques, kuils, community halls and malls**. **Next steps:** record 20+ more cycles to measure our false-alarm rate, then pilot on real township equipment.

7. EXPECTED IMPACT, SCALABILITY & COMMERCIAL ROI

Example (our estimate, to be confirmed in the pilot):
- One 5 kW commercial AC running 10 hours a day uses about 18,250 kWh a year. If a worn part makes it use just **10% more**, that is ~1,825 kWh — **about RM 900 a year** at ~RM 0.50/kWh.
- A pilot kit costs about **RM 170**, so catching one problem like this **pays for the pod within a few months**.
- Catching a failing part early can also **avoid replacing a whole compressor**, which costs thousands of ringgit.

Township Quality of Life & ESG Impact:

- **Less Disruption:** Repairs can be planned ahead, so cooling stays on during Friday prayers, festivals and busy weekends.
- **Less Waste & Lower Emissions:** Fixing one small part instead of replacing a whole unit keeps machines running longer, cuts scrap metal and saves electricity.